# From "Direct" to "Indirect" Generative Models

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**Abstract**

Many generative models studied in modern machine learning are based around specifying an interesting 'direct' generative process for the object of interest $x$. Some recent methods deviate from this perspective by acting 'indirectly', instead focusing on generating indirect measurements $y$ of $x$, and then finally reconstructing $x$ conditional on $y$. In this note, I describe a common basis for some of these methods, grounded in a simple calibration property of Bayesian inference, and show how these models can be obtained by simple manipulation of joint distributions.

## 1 Generative Modelling, Direct and Indirect

In machine learning, *generative modeling* is a modern twist on the classical task of density estimation: given samples $x^1, \cdots, x^N$ which are assumed to be drawn from a common law $p_\star$ on a space $\mathcal{X}$, construct a measure-valued estimator $\hat{p}$ of $p_\star$. Conventional density estimators provide $\hat{p}$ as a simple function which can be explicitly evaluated pointwise, and is evaluated on those terms. By contrast, contemporary generative modeling focuses on other functionalities, with a particular focus on tractability of simulation from $\hat{p}$.

Some classes of generative model have a flavour of 'direct' generation; the model is specified through an explicit algorithm which constructs a sample $x \sim \hat{p}$. Autoregressive models build the sample one coordinate at a time ('ancestrally'), whereas other models (e.g. generative adversarial networks, normalising flows, diffusion models) generative the sample by passing a 'boring' random variable (e.g. uniform, Gaussian) through an interesting dynamical system to output some $x$ which is more interesting. To some extent, the conceptual excitement of 'direct' generative models then lies in cleverly parametrising such dynamical systems to endow them with both flexibility and tractability.

By contrast, some other model classes are instead 'indirect' in character, in that first, some nontrivial auxiliary random variable $y$ is formed, and is then used as a basis for guiding the generation of the final sample $x$. Sometimes, this $y$ carries the interpretation of a 'representation' or 'encoding' of $x$, which is then ultimately 'decoded'. Other times, $y$ instead represents an indirect observation of (the as-yet uninstantiated) $x$, and the ultimate generation of $x$ given $y$ carries the interpretation of signal inference. There is little mathematical distinction between these two possibilities, in as much as indirect observations of a signal can be viewed as a lossy encoding (or conversely, that any parameter is identified by a sufficiently informative collection of observations of it).

Note briefly that even the distinction between 'direct' and 'indirect' generation is largely a matter of perspective; I will only use these terms to the extent that they can streamline understanding and common threads between different approaches. The purpose of this note is then to detail how several 'indirect' approaches to generative modeling can be motivated by basic properties of the Bayesian paradigm, and that different methodologies can be mapped onto different interpretations of a common joint distribution.

## 2 Bayesian Calibration

Consider Bayesian inference of a parameter $x_\star$ which is equipped with prior law $\pi(x)$, and which is observed indirectly through the signal $y \sim \ell(x, y)$. In the Bayesian paradigm, decisions are made on the basis of the posterior law

$$\Pi(x \mid y) = \pi(x) \cdot \ell(x, y) \cdot (\pi\ell)(y)^{-1},$$

where $(\pi\ell)(y) = \int_{\mathcal{X}} \pi(x) \cdot \ell(x, y)\, \mathrm{d}x$ is the prior predictive density of $y$. Consider then the joint law of the true parameter $x_\star$, the observation $y$, and a posterior reconstruction $\hat{x}$, i.e.

$$\mathsf{Joint}(x_\star, y, \hat{x}) = \pi(x_\star) \cdot \ell(x_\star, y) \cdot \Pi(\hat{x} \mid y).$$

A re-factorisation of the terms yields the equivalent representation

$$\mathsf{Joint}\left(x_\star, y, \hat{x}\right) = \pi\left(x_\star\right) \cdot \ell\left(x_\star, y\right) \cdot \pi\left(\hat{x}\right) \cdot \ell\left(\hat{x}, y\right) \cdot \left(\pi\ell\right)\left(y\right)^{-1},$$

which is readily seen to be symmetric with respect to $(x_\star, \hat{x})$. In statistical physics, a version of this observation is known as the 'Nishimori identity'.

Examining this statement, one can deduce that for a well-specified Bayesian model, a perfect sample from the posterior law is **marginally** drawn from the prior. That is, integrating out $(x_\star, y)$, the law of $\hat{x}$ is precisely $\pi$. This is a form of calibration statement for the idealised Bayesian statistician: averaged over the probable data $y$, our reconstructions $\hat{x}$ of the true parameter $x_\star$ will reproduce the prior law $\pi$.

Speaking impractically, this suggests a strange algorithm for sampling from $\pi$:

1. Sample a true parameter $x_\star \sim \pi$.
2. Generate synthetic data $y \sim \ell\left(x_\star, \cdot\right)$.
3. Reconstruct $\hat{x} \sim \Pi\left(\cdot \mid y\right)$,

and retain $\hat{x}$ as the sample. This is not a particularly sensible algorithm on its face: by step 1, we have already sampled from the distribution of interest, so there is little need to stick around for the subsequent steps. Still, we will see that this general strategy has some interesting features.

In subsequent developments, it will be useful to consider the setting of sequential Bayesian inference, i.e. for $1 \leqslant t \leqslant T$, one observes $y_t \sim \ell_t\left(x, \cdot\right)$. I will write $\Pi_t\left(\cdot \mid y_{\leqslant t}\right)$ for the corresponding intermediate posterior laws. The joint law of all generated variables (which I denote by $\mathbf{P}$) then corresponds to the procedure

1. Sample a true parameter $x_\star \sim \pi$.
2. For $1 \leqslant t \leqslant T$, generate synthetic data $y_t \sim \ell_t\left(x_\star, \cdot\right)$.
3. Reconstruct $\hat{x} \sim \Pi_T\left(\cdot \mid y_{\leqslant T}\right)$,

noting that the same symmetry property continues to hold for $(x_\star, \hat{x})$ under this law.

# 3 Alternative Model Factorisations

In this section, I will detail some alternative factorisations for the joint law $\mathbf{P}\left(x_\star, y_{1:T}, \hat{x}\right)$. Throughout, I will be describing (conceptual) algorithms which generate variables following the law of $\mathbf{P}$, though the causal structure of how these variables will vary from the original setting. In a way, this is what allows for the creativity in these models.

## 3.1 Prequential Factorisation

In the preceding section, we implicitly presented our joint law in line with the usual causal presentation of Bayesian inference, i.e. one draws the true parameter $x_\star$, then simulates the data $y$, and then forms a reconstruction of the parameter $\hat{x}$. Still, when speaking only about the joint law in abstract, one is free to express the model in a data-first manner.

That is, defining $\lambda_t$ to be the predictive law of the $t^{\text{th}}$ observation, conditional on all data $y$ observed at times $s < t$, one can write down an equivalent generative procedure:

1. For $1 \leqslant t \leqslant T$, generate synthetic data $y_t \sim \lambda_t\left(\cdot \mid y_{<t}\right)$.
2. Reconstruct $x_\star, \hat{x} \sim \Pi_T\left(\cdot \mid y_{\leqslant T}\right)$, where the draws are independent.

Outside of a narrow class of special priors and observation models, the form of $\lambda_t$ is not generally available analytically. Nevertheless, if one could genuinely sample from these predictive laws, one might have a reasonable chance at using this method to sample from $\pi$. In particular, for large $T$ and informative $\ell$, one expects that the posterior law $\Pi_T\left(\cdot \mid y_{\leqslant T}\right)$ will be well-concentrated, and so (approximate) reconstruction of $x_\star, \hat{x}$ from this law may be manageable.

### 3.2 Prequential Factorisation with Posterior Unfolding

The implicit nature of $\lambda_t$ can be demystified somewhat by noting that

$$\begin{aligned}\mathbf{P}\left(y_t \mid y_{<t}\right) &= \int \mathbf{P}\left(x, y_t \mid y_{<t}\right) \mathrm{d}x \\ &= \int \Pi_{t-1}\left(x \mid y_{<t}\right) \cdot \ell_t\left(x, y_t\right) \mathrm{d}x,\end{aligned}$$

that is, one-step-ahead prediction at time $t$ can be realised as a Markov composition of i) reconstructing the true parameter through the intermediate posterior $\Pi_{t-1}\left(\cdot \mid y_{<t}\right)$ and ii) using this estimator to generate observations according to the original observation model $\ell_t$. Interpreted algorithmically, this then yields an expanded joint law which also admits $\mathbf{P}$ as its marginal:

1. For $1 \leqslant t \leqslant T$,
   (a) Form the intermediate reconstruction $\bar{x}_t \sim \Pi_{t-1}\left(\cdot \mid y_{<t}\right)$.
   (b) Generate synthetic data $y_t \sim \ell_t\left(\bar{x}_t, \cdot\right)$.
2. Reconstruct $x_\star, \hat{x} \sim \Pi_T\left(\cdot \mid y_{\leqslant T}\right)$, where the draws are independent.

Relative to the original prequential factorisation, this procedure essentially exchanges the task of predictive simulation (i.e. drawing from $y_t \sim \lambda_t\left(\cdot \mid y_{<t}\right)$) with posterior simulation (i.e. drawing from $\bar{x}_t \sim \Pi_{t-1}\left(\cdot \mid y_{<t}\right)$), assuming that the basic observation model $\ell_t$ is chosen so that it admits cheap simulation.

## 4 From Model Factorisations to Generative Modeling

The preceding derivations are phrased in a way which suggests that they might be appropriate for developing sampling algorithms for inference in Bayesian models. There is a widely-understood (though rarely-documented) phenomenon that various procedures for approximating an unnormalised probability measure can be converted into methods for { density estimation / generative modeling }. In this section, I will describe how to enact this conversion in the context of our joint model from Sections 2 and 3.

In essence, the key step will be to re-visit the previous factorisations while taking $\pi = p_\star$. In particular, with the ideal 'forward' model as

1. Sample a true parameter $x_\star \sim p_\star$.
2. For $1 \leqslant t \leqslant T$, generate synthetic data $y_t \sim \ell_t\left(x_\star, \cdot\right)$.
3. Reconstruct $\hat{x} \sim \Pi_T\left(\cdot \mid y_{\leqslant T}\right)$,

one can analyse the alternative factorisations corresponding to this model, describe suitable approximations to them, and use these approximation to define the generative model $\hat{p} = \mathrm{Law}\left(\hat{x}\right)$. In what follows, assume throughout that the observation models $\ell = \{\ell_t : 1 \leqslant t \leqslant T\}$ are fixed a priori.

### 4.1 From Prequential Factorisation to Stochastic Localisation and Martingale Posteriors

Recall the model in Subsection 3.1:

1. For $1 \leqslant t \leqslant T$, generate synthetic data $y_t \sim \lambda_t\left(\cdot \mid y_{<t}\right)$.
2. Reconstruct $x_\star, \hat{x} \sim \Pi_T\left(\cdot \mid y_{\leqslant T}\right)$, where the draws are independent.

The main task in implementing a practical version of this procedure is then to describe explicit approximations $\hat{\lambda} = \left\{\hat{\lambda}_t : 1 \leqslant t \leqslant T\right\}$ to the predictive law of the data, and to provide a route towards fitting such an approximation in practice.

While this is expected to be challenging for general $\ell$, a remarkable observation of the past few years has been that for highly-structured choices of observation model, the predictive laws $\lambda$ are often concisely expressible. Section 4 of [3] details numerous examples to this effect, also describing a connection to recent developments in the growing field of diffusion-based generative modeling.

In particular, in suitable limiting regimes, one finds that the $\lambda_t$ lie in simple parametric families of distributions, with parameters expressible in terms of conceptually simple quantities, typically a conditional expectation of some simple function under the joint law of $(x_\star, y_{<t})$, e.g.

$$\begin{aligned}\lambda_t\left(y_t \mid y_{<t}\right) &= \mathsf{ParametricFamily}\left(y_t; \text{parameter} = \eta_t\left(y_{<t}\right)\right)\\ \eta_t\left(y_{<t}\right) &= \mathbf{E}\left[h_t\left(X_\star\right) \mid Y_{<t} = y_{<t}\right]\end{aligned}$$

for some explicit function $h_t$. This then enables the straightforward development of computational schemes for estimating these relevant quantities, as conditional expectations can readily be estimated by simple regression procedures. (This is not to say that they will always work well - it is still a regression problem, potentially in a high-dimensional space - but conceptually, nothing deep needs to occur)

An alternative approach is taken in [1], wherein the law $\lambda_t$ is modeled directly through a copula-based procedure; the focus there is on posterior inference rather than generative modeling.

## 4.2 From Unfolded Prequential Factorisation to Bayesian Flow Networks

Recall the model in Subsection 3.2:

1. For $1 \leqslant t \leqslant T$,
   (a) Form the intermediate reconstruction $\bar{x}_t \sim \Pi_{t-1}\left(\cdot \mid y_{<t}\right)$.
   (b) Generate synthetic data $y_t \sim \ell_t\left(\bar{x}_t, \cdot\right)$.
2. Reconstruct $x_\star, \hat{x} \sim \Pi_T\left(\cdot \mid y_{\leqslant T}\right)$, where the draws are independent.

The main task in implementing a practical version of this procedure is then to describe explicit approximations $\hat{\Pi} = \left\{\hat{\Pi}_t : 0 \leqslant t < T\right\}$ to the intermediate posterior distributions, and to provide a route towards fitting such an approximation in practice.

The approach of Bayesian Flow Networks [2] is to model $\hat{\Pi}$ as an element of some parametric family, and obtain its parameters over time by analogy with Bayesian updating in conjugate models. The setup is roughly as follows:

1. Posit an auxiliary space $\mathcal{T}$ which will contain summary statistics of intermediate datasets, and a reference point $\tau_0 \in \mathcal{T}$.
2. Specify a collection of assimilation mappings $\{\mathsf{Update}_t : 1 \leqslant t \leqslant T\} : \mathcal{T} \times \mathcal{Y} \to \mathcal{T}$.
3. Select a parametric family of approximate posterior distributions, $\mathcal{Q} = \{q\left(x; \phi\right) : \phi \in \Phi\} \subseteq \mathcal{P}\left(\mathcal{X}\right)$.
4. Specify a collection of summary-to-parameter mappings $\{\varphi_t : 0 \leqslant t \leqslant T\} : \mathcal{T} \to \Phi$.

Some brief comments on notation: I write $\mathcal{T}$ for the space of summary statistics to match with classical notation for (sufficient) statistics in exponential families; this matches with their $\theta$ (which is arguably more evocative of 'parameters'). I use $\Phi$ to denote the parameter space of the approximating family (by analogy with many works in variational inference), and my map $\varphi_t$ matches with their $\Psi\left(t, \cdot\right)$.

In terms of intuition, it is useful to consider briefly the task of Bayesian inference in a conjugate exponential family, wherein one need only i) remember the prior hyperparameters, and ii) track the sum of the sufficient statistics across observations. In this setting, the $\mathsf{Update}$ maps can simply correspond to incrementing the cumulative sum of the sufficient statistics with the most recent observation, i.e. $\mathsf{Update}_t\left(\tau, y_t\right) = \tau + T\left(y_t\right)$, where $T$ is the sufficient statistic of the exponential family.

With this setup established, we can recursively form approximate posteriors as follows (suppressing the qualifier 'approximate' throughout for conciseness):

1. At initialisation, $\tau_0$ is given; compute $\phi_0 = \varphi_0\left(\tau_0\right)$.
2. For $1 \leqslant t \leqslant T$,
   (a) Form the intermediate posterior $\hat{\Pi}_{t-1}\left(x \mid y_{<t}\right) := q\left(x \mid \phi_{t-1}\right)$.
   (b) Form the intermediate reconstruction $\bar{x}_t \sim \hat{\Pi}_{t-1}\left(x \mid y_{<t}\right)$.
   (c) Generate synthetic data $y_t \sim \ell_t\left(\bar{x}_t, \cdot\right)$.
   (d) Update the summary statistic $\tau_t = \mathsf{Update}_t\left(\tau_{t-1}, y_t\right)$.

(e) Update the posterior parameter $\phi_t = \varphi_t(\tau_t)$.

3. Form the final posterior $\hat{\Pi}_T(x \mid y_{\leqslant T}) := q(x \mid \phi_T)$.

4. Reconstruct $x_\star, \hat{x} \sim \hat{\Pi}_T(\cdot \mid y_{\leqslant T})$, where the draws are independent.

At present, this simply parametrises a generative process $\hat{p} = \mathrm{Law}(\hat{x})$, without explaining how one might fit the various free parameters. The approach of [2] is then to treat $\hat{p}$ as a latent variable model, with $y_{1:T}$ as latent variables (the model is designed so that $\bar{x}_{1:T}$ can be marginalised out analytically) and then fit the model by a form of Variational EM, as is done for variational autoencoders. They use a relatively simple variational family, taking the form $r(y_{1:T} \mid x_\star) = \prod_{1 \leqslant t \leqslant T} \ell_t(\mathrm{d}y_t \mid x_\star)$.

There are a couple of other design choices in the BFN paper which can be commented on quickly: i) they take the observation model $\ell$ to factorise across coordinates, i.e. $\ell_t(x, y) = \prod_{i \in [\![d]\!]} \ell_t(x^i, y^i)$, ii) they take the approximate posterior model to factorise across coordinates, compatibly with the summary-to-parameter map $\varphi$, i.e. $q(x \mid \varphi_t(\tau)) = \prod_{i \in [\![d]\!]} q(x^i \mid \varphi_t^i(\tau))$, iii) the $\mathsf{Update}$ functions are constrained to have a sort of semigroup / additivity property, related to an 'accuracy' hyperparameter $\alpha_t$ associated with the informativeness of the observation $\ell_t$, and iv) they construct $\mathcal{Q}$ to be compatible with $\ell$ so that the composite $y \mapsto \int q(x \mid \phi) \cdot \ell(x, y)\, \mathrm{d}x$ can be treated in closed form. They also describe an 'input distribution' which subsequently seems to disappear from use.

# 5 Conclusion

In this note, I have reviewed a simple identity concerning Bayesian inference, and have shown how it can be used to describe a number of strange procedures for sampling from a prior distribution. I have then explained how the underlying ideas can be adapted to the context of generative modeling, and drawn connections to existing works.

# References


[1] Edwin Fong, Chris Holmes, and Stephen G Walker. Martingale posterior distributions. *arXiv preprint arXiv:2103.15671*, 2021.

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